

14.4 The correlation coefficient and Fisher's Z

With a single regressor, the multiple correlation reduces to the ordinary correlation ρ of the pair (X, Y) , and R^2 to the square of the sample correlation

$$r = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2}}.$$

Keeping the sign, this section develops two routes to inference about ρ : the exact distribution of r , and Fisher's transformation to a nearly normal variable whose variance does not depend on ρ . Throughout, (X_i, Y_i) , $i = 1, \dots, n$, are independent draws from a bivariate normal distribution with correlation $|\rho| < 1$, and $n \geq 3$.

14.4.1 The exact distribution of r

Proposition 14.4.1 — Exact law of the sample correlation

Let $T = r\sqrt{n-2}/\sqrt{1-r^2}$ and $\delta = \rho/\sqrt{1-\rho^2}$.

- (a) Given the X values, $T \sim t(n-2, \delta\sqrt{V})$, where $V = \sum_i (X_i - \bar{X})^2 / \sigma_X^2$.
- (b) $V \sim \chi^2(n-1)$, and for $-1 < c < 1$,

$$\Pr(r \leq c) = \mathbb{E} \left[\Pr\{t(n-2, \delta\sqrt{V}) \leq c\sqrt{n-2}/\sqrt{1-c^2}\} \right]. \quad (14.4.1)$$

- (c) If $\rho = 0$, then $T \sim t(n-2)$ and $r^2 \sim \text{Beta}(\frac{1}{2}, \frac{n-2}{2})$, independently of the X values.
- (d) The law of r is stochastically increasing in ρ : for each $c \in (-1, 1)$, $\Pr(r \leq c)$ is a strictly decreasing function of ρ .

Proof.

- (a) Let $\hat{\beta}_1$ be the least squares slope of Y on X and t_1 its t statistic for the value zero. By Theorem 9.5.4(c) (see also Exercise (B3)), with only the intercept as the other column, $r^2 = t_1^2 / (t_1^2 + n - 2)$, and t_1 has the sign of $\hat{\beta}_1$, which is the sign of r . Solving, $t_1 = T$. Given the X values the pairs follow a normal simple regression with slope $\beta_1 = \rho\sigma_Y/\sigma_X$ and error variance $\sigma^2 = \sigma_Y^2(1-\rho^2)$ (Eq. 3.5.2), so by Corollary 7.5.2 with $d = 0$, T is noncentral t with $n-2$ degrees of freedom and noncentrality $\beta_1\sqrt{S_{xx}}/\sigma = \rho\sqrt{S_{xx}}/(\sigma_X\sqrt{1-\rho^2}) = \delta\sqrt{V}$.
- (b) $V \sim \chi^2(n-1)$ by Corollary 4.4.5. Since T is an increasing function of r , $r \leq c$ iff $T \leq c\sqrt{n-2}/\sqrt{1-c^2}$, and Eq. 14.4.1 is the expectation of the conditional probability.
- (c) With $\delta = 0$ the conditional law is $t(n-2)$ for every design; apply Lemma 14.1.1. Then $r^2 = T^2/(T^2 + n - 2)$ is the $k = 1$ case of Theorem 14.3.2(b).
- (d) Write a $t(\nu, \theta)$ variable as $(Z + \theta)/\sqrt{W/\nu}$ (Definition 4.1.3). For $\theta_1 < \theta_2$ and any u , $\Pr\{t(\nu, \theta_1) \leq u\} - \Pr\{t(\nu, \theta_2) \leq u\} = \Pr\{u\sqrt{W/\nu} - \theta_2 < Z \leq u\sqrt{W/\nu} - \theta_1\} > 0$. Since $\delta = \rho/\sqrt{1-\rho^2}$ is strictly increasing in ρ , for every c and every $V > 0$ the conditional probability in Eq. 14.4.1 strictly decreases in ρ , and so does its

expectation.

■

Fisher (1915) found the density of r as an infinite series; for computation the mixture Eq. 14.4.1, an average of noncentral t probabilities over quantiles of V , is enough. Part (c) is the familiar test of $\rho = 0$, the t test of a zero slope, symmetric in the two variables. Part (d) makes exact intervals possible, by the inversion used for ρ^2 in Section 14.3.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, stats

def r_cdf(c, rho, n, m=4000):
    """Exact P(r <= c) for n iid bivariate normal pairs with correlation rho."""
    if c < 0:
        # law of -r is that for
        -rho
        return 1 - r_cdf(-c, -rho, n, m)
    t = c * np.sqrt(n - 2) / np.sqrt(1 - c ** 2)
    # increasing in c
    if rho == 0:
        return stats.t.cdf(t, n - 2)
    V = stats.chi2.ppf((np.arange(m) + 0.5) / m, n - 1)
    # equal-probability
    nodes
    delta = rho / np.sqrt(1 - rho ** 2)
    return np.mean(stats.nct.cdf(t, n - 2, delta * np.sqrt(V)))
```

[Run this code online](#)

When $\rho \neq 0$ the distribution of r is skewed towards zero, because r cannot pass ± 1 . For $\rho = 0.8$ and $n = 15$, 200000 simulated samples give a mean of 0.7886, a median of 0.811, a skewness of -1.40 and a variance of 0.01195 (panel (a) of Figure 14.4.1). The variance depends strongly on ρ : the standard deviation of r is 0.267 at $\rho = 0$ and 0.032 at $\rho = 0.95$ (panel (c)). An interval $r \pm \text{constant} \times \text{standard error}$ would inherit both problems.

14.4.2 Fisher's transformation

Fisher (1921) proposed to work instead with

$$z = \tanh^{-1} r = \frac{1}{2} \log \frac{1+r}{1-r}, \quad \zeta = \tanh^{-1} \rho. \quad (14.4.2)$$

To see why this helps we need the large-sample behaviour of r , and for that three standard tools from probability theory, used here without proof (see van der Vaart 1998, chapters 2 and 3).

Remark (Three limit theorems).

- (i) *Central limit theorem.* If $\mathbf{W}_1, \mathbf{W}_2, \dots$ are independent copies of a random vector with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Gamma}$, then $\sqrt{n}(\bar{\mathbf{W}}_n - \boldsymbol{\mu}) \rightarrow N(\mathbf{0}, \boldsymbol{\Gamma})$ in distribution.
- (ii) *Slutsky's lemma.* If $\mathbf{U}_n \rightarrow \mathbf{U}$ in distribution and $\mathbf{V}_n \rightarrow \mathbf{c}$ in probability, then $\mathbf{U}_n + \mathbf{V}_n \rightarrow \mathbf{U} + \mathbf{c}$, and $V_n U_n \rightarrow cU$ for scalars.
- (iii) *Delta method.* If $\sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta}) \rightarrow$

$N(\mathbf{0}, \mathbf{\Gamma})$ and g is differentiable at $\boldsymbol{\theta}$ with gradient ∇g , then $\sqrt{n}(g(\mathbf{T}_n) - g(\boldsymbol{\theta})) \rightarrow N(0, \nabla g^\top \mathbf{\Gamma} \nabla g)$.

The limiting variance of r needs fourth moments of the bivariate normal distribution.

Lemma 14.4.2 — Fourth moments of a standard bivariate normal pair

Let (X, Y) be bivariate normal with means 0, variances 1 and correlation ρ . Then $E X^4 = 3$, $E X^3 Y = 3\rho$ and $E X^2 Y^2 = 1 + 2\rho^2$. Consequently the covariance matrix of (X^2, Y^2, XY) is

$$\mathbf{\Gamma} = \begin{pmatrix} 2 & 2\rho^2 & 2\rho \\ 2\rho^2 & 2 & 2\rho \\ 2\rho & 2\rho & 1 + \rho^2 \end{pmatrix}.$$

Proof. Write $Y = \rho X + \sqrt{1 - \rho^2} U$ with $U \sim N(0, 1)$ independent of X ; the pair has the right means, variances and covariance, and it is normal by Theorem 3.3.1. Using $E X^2 = 1$, $E X^4 = 3$ (the fourth moment of a standard normal) and the vanishing of odd moments, $E X^3 Y = \rho E X^4 = 3\rho$ and $E X^2 Y^2 = \rho^2 E X^4 + (1 - \rho^2) E X^2 E U^2 = 3\rho^2 + 1 - \rho^2 = 1 + 2\rho^2$. The covariances follow by subtracting products of means, $E X^2 = E Y^2 = 1$ and $E XY = \rho$: for instance $\text{Var}(XY) = 1 + 2\rho^2 - \rho^2$ and $\text{Cov}(X^2, XY) = E X^3 Y - \rho = 2\rho$. ■

Theorem 14.4.3 — Fisher's z transformation

As $n \rightarrow \infty$,

$$\sqrt{n}(r - \rho) \rightarrow N(0, (1 - \rho^2)^2) \quad \text{and} \quad \sqrt{n}(z - \zeta) \rightarrow N(0, 1)$$

in distribution. The same limits hold with $\sqrt{n-3}$ in place of $\sqrt{n}$. The limiting variance of z does not depend on ρ : $\tanh^{-1}$ is a variance-stabilizing transformation for r .

Proof. The correlation is unchanged by the maps $X \mapsto a + bX$, $Y \mapsto c + dY$ with $b, d > 0$, so we may assume $E X = E Y = 0$ and $\text{Var } X = \text{Var } Y = 1$. Let $\bar{\mathbf{W}}_n$ be the average of $\mathbf{W}_i = (X_i^2, Y_i^2, X_i Y_i)$, whose mean is $\boldsymbol{\theta} = (1, 1, \rho)$ and whose covariance is $\mathbf{\Gamma}$ of Lemma 14.4.2. With divisor n , the sample variances and covariance are $\bar{\mathbf{W}}_n - (\bar{X}^2, \bar{Y}^2, \bar{X}\bar{Y})$. Since $\sqrt{n} \bar{X}$ converges in distribution and $\bar{X} \rightarrow 0$ in probability, $\sqrt{n} \bar{X}^2 \rightarrow 0$ in probability, and similarly for the other two products, so by Slutsky's lemma the centring does not affect the limit. The divisor cancels in $r = g(\text{sample moments})$, where $g(a, b, c) = c/\sqrt{ab}$, and $\nabla g(\boldsymbol{\theta}) = (-\rho/2, -\rho/2, 1)$. By the central limit theorem and the delta method, $\sqrt{n}(r - \rho)$ is asymptotically normal with variance

$$\nabla g^\top \mathbf{\Gamma} \nabla g = \frac{\rho^2}{4}(2+2+4\rho^2) - 2 \cdot \frac{\rho}{2}(2\rho+2\rho) + (1+\rho^2) = \rho^2 + \rho^4 - 4\rho^2 + 1 + \rho^2 = (1 - \rho^2)^2.$$

The derivative of $\tanh^{-1}$ at ρ is $1/(1 - \rho^2)$, so a second application of the delta method gives variance 1 for z . Finally $\sqrt{n-3}/\sqrt{n} \rightarrow 1$, and Slutsky's lemma allows the substitution.

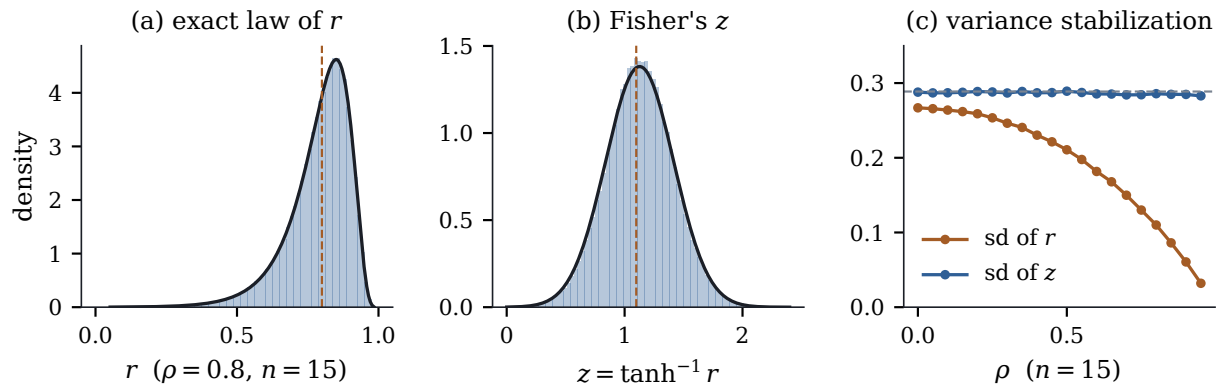


Figure 14.4.1: The sample correlation of $n = 15$ normal pairs. (a) r for $\rho = 0.8$, with the exact density, obtained by differentiating Eq. 14.4.1. (b) Fisher's z , with the normal approximation. (c) The standard deviations of r and z across ρ ; dashed: $1/\sqrt{n-3}$.

The transformation solves $g'(\rho) = 1/(1 - \rho^2)$, the condition for a limiting standard deviation of one (Exercise (B1)). Why the approximation is so good for small n is another matter: Fisher (1921) showed that the mean of z exceeds ζ by about $\rho/(2(n-1))$ and that its variance is close to $1/(n-3)$, and the distribution of z is nearly symmetric even when that of r is strongly skewed. In the simulation with $\rho = 0.8$ and $n = 15$, z has mean 1.1295 against $\zeta = 1.0986$ and $\zeta + \rho/(2(n-1)) = 1.1272$, variance 0.0811 against $1/(n-3) = 0.0833$, and skewness 0.01. Its standard deviation is 0.288 at $\rho = 0$ and 0.283 at $\rho = 0.95$ (panel (c)).

14.4.3 Intervals and tests for a correlation

The approximation $z \approx N(\zeta, 1/(n-3))$ gives the interval

$$\tanh\left(z - \frac{z_{\alpha/2}}{\sqrt{n-3}}\right) \leq \rho \leq \tanh\left(z + \frac{z_{\alpha/2}}{\sqrt{n-3}}\right), \quad (14.4.3)$$

with $z_{\alpha/2}$ the upper $\alpha/2$ point of $N(0,1)$. It is asymmetric about r , as it should be, and it never leaves $(-1,1)$. The exact interval inverts Eq. 14.4.1, using part (d) of Proposition 14.4.1 exactly as for ρ^2 : its limits are the values of ρ at which the observed r is the 97.5% and the 2.5% point.

How good is the approximation? For $n = 10$ pairs, the coverage of the 95% interval Eq. 14.4.3 over 200000 samples is 0.949, 0.949 and 0.952 at $\rho = 0, 0.5$ and 0.9 . The interval $r \pm 1.96(1 - r^2)/\sqrt{n}$, which uses the asymptotic standard error of Theorem 14.4.3 directly, covers only 0.839, 0.838 and 0.834.

```
def sample_r(rho, n, reps, rng):
    x = rng.normal(size=(reps, n))
    y = rho * x + np.sqrt(1 - rho ** 2) * rng.normal(size=(reps, n))
    xc = x - x.mean(axis=1, keepdims=True)
    yc = y - y.mean(axis=1, keepdims=True)
    return np.sum(xc * yc, axis=1) / np.sqrt(np.sum(xc ** 2, axis=1) * np.sum(yc **
2, axis=1))

rng = np.random.default_rng(1443)
```

```

n_c, reps_c, zq = 10, 200_000, stats.norm.ppf(0.975)
for rho_c in (0.0, 0.5, 0.9):
    r_c = sample_r(rho_c, n_c, reps_c, rng)
    z_c = np.arctanh(r_c)
    fisher = np.abs(z_c - np.arctanh(rho_c)) <= zq / np.sqrt(n_c - 3)
    naive = np.abs(r_c - rho_c) <= zq * (1 - r_c ** 2) / np.sqrt(n_c)
    print(f"rho = {rho_c}: Fisher z covers {fisher.mean():.4f}, naive covers {naive.
          mean():.4f}")

```

[Run this code online](#)

Because z has an approximately known variance, it also compares correlations. With independent samples of n_1 and n_2 cases, the hypothesis $\rho_1 = \rho_2$ is tested by referring

$$\frac{z_1 - z_2}{\sqrt{1/(n_1 - 3) + 1/(n_2 - 3)}}$$

to the standard normal distribution. The t statistic of [Proposition 14.4.1\(c\)](#) remains the exact test of $\rho = 0$; for any other null value ρ_0 , either $(z - \tanh^{-1} \rho_0)\sqrt{n - 3}$ or the exact law [Eq. 14.4.1](#) is used.

Example. (Poverty and graduation)

Across the 50 states, the correlation between the poverty rate and the high-school graduation rate is $r = -0.7613$. Its t statistic for $\rho = 0$ is -8.13 on 48 degrees of freedom, far out in the tail. The Fisher interval [Eq. 14.4.3](#) is $(-0.858, -0.613)$ and the exact interval is $(-0.855, -0.609)$, almost the same, and both are noticeably asymmetric about r .

```

data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
x, y = data["poverty"].to_numpy(), data["hs_grad"].to_numpy()
n_s = len(x)
r_s = np.corrcoef(x, y)[0, 1]
t_s = r_s * np.sqrt(n_s - 2) / np.sqrt(1 - r_s ** 2)
lo_z, hi_z = np.tanh(np.arctanh(r_s) + np.array([-1, 1]) * zq / np.sqrt(n_s - 3))
# exact interval: the rho at which the observed r sits at the 97.5% and 2.5% points
lo_x = optimize.brentq(lambda rh: r_cdf(r_s, rh, n_s) - 0.975, -0.99, 0.99)
hi_x = optimize.brentq(lambda rh: r_cdf(r_s, rh, n_s) - 0.025, -0.99, 0.99)
print(f"r = {r_s:.4f}, t = {t_s:.2f} on {n_s - 2} df")
print(f"Fisher z interval ({lo_z:.3f}, {hi_z:.3f}); exact interval ({lo_x:.3f}, {
      hi_x:.3f})")

```

[Run this code online](#)

Warning. Everything in this section uses bivariate normality, and not only through the conditional linear model. The asymptotic variance $(1 - \rho^2)^2/n$ comes from the fourth moments of [Lemma 14.4.2](#). For other distributions it can be quite different, and then neither z nor the exact interval is reliable ([Exercise \(B2\)](#)). The t test of $\rho = 0$ is more robust: by Theorem 14.1.2 it needs only that Y given X follow a normal linear model with constant variance, and when X and Y are independent this holds as soon as Y is normal.

14.4.4 Exercises

14.4.4.1 A. Check your understanding

Exercise A1.

A sample of $n = 28$ pairs has $r = 0.62$. Compute the 95% Fisher interval for ρ .

Solution. $z = \tanh^{-1} 0.62 = 0.7250$ and $1.96/\sqrt{25} = 0.392$, so ζ lies in $(0.333, 1.117)$, and ρ in $(\tanh 0.333, \tanh 1.117) = (0.321, 0.806)$. The interval is longer below r than above.

Exercise A2.

Two independent samples give $r_1 = 0.45$ with $n_1 = 60$ and $r_2 = 0.20$ with $n_2 = 80$. Test $\rho_1 = \rho_2$ at level 0.05.

14.4.4.2 B. Practice

Exercise B1.

Suppose $\sqrt{n}(T_n - \theta) \rightarrow N(0, s(\theta)^2)$ for every θ in an interval, with s positive and continuous. Show that if $g'(\theta) = 1/s(\theta)$, then $\sqrt{n}(g(T_n) - g(\theta)) \rightarrow N(0, 1)$. Solve $g'(\rho) = 1/(1 - \rho^2)$ and recover [Eq. 14.4.2](#).

Solution. By the delta method the limiting variance is $g'(\theta)^2 s(\theta)^2 = 1$. With partial fractions, $1/(1 - \rho^2) = \frac{1}{2}[1/(1 + \rho) + 1/(1 - \rho)]$, whose antiderivative vanishing at zero is $\frac{1}{2}[\log(1 + \rho) - \log(1 - \rho)] = \tanh^{-1} \rho$.

Exercise B2.

Let X and U be independent standard normal variables and $Y = XU$. Show that X and Y are uncorrelated but not independent, and that $\sqrt{n}r \rightarrow N(0, 3)$. What happens to the level of the test that rejects $\rho = 0$ when $|T| > t_{n-2, 0.025}$ in large samples? Which assumption of Theorem 14.1.2 fails?

Solution. $EXY = EX^2EU = 0$, but $E(X^2Y^2) = EX^4EU^2 = 3 \neq EX^2EY^2 = 1$, so they are dependent. Repeat the proof of [Theorem 14.4.3](#) at $\rho = 0$: the gradient is $(0, 0, 1)$, so the limiting variance of $\sqrt{n}r$ is $\text{Var}(XY)/(\text{Var } X \text{Var } Y) = 3$. Since $T \approx \sqrt{n}r$ for large n , the test rejects with probability tending to $2\{1 - \Phi(1.96/\sqrt{3})\} = 0.258$, not 0.05. Given X , Y is normal with mean zero but variance X^2 : the errors are not homoscedastic, so the conditional model of Definition 14.1.1 fails.

14.4.4.3 C. Going deeper

Exercise C1.

For a general pair (X, Y) with finite fourth moments, standardized to means 0 and vari-

ances 1, show that $\sqrt{n}(r - \rho)$ is asymptotically normal with variance $(1 + \frac{\rho^2}{2}) E X^2 Y^2 + \frac{\rho^2}{4} (E X^4 + E Y^4) - \rho(E X^3 Y + E X Y^3)$. Check that it reduces to $(1 - \rho^2)^2$ under normality, and explain why no transformation of r stabilizes the variance across all distributions.